BENSON LIN

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Open to relocation anywhere in the U.S.

EDUCATION

Worcester Polytechnic Institute May 2026
Master of Science in Computer Science Worcester, MA

Worcester Polytechnic Institute May 2026
Bachelor of Science in Computer Science (GPA: 3.56/4.0) Worcester, MA

SKILLS

Languages : Python, Java, SQL, TypeScript, JavaScript
Backend : FastAPI, Spring Boot, REST APIs, WebSockets, SQLAlchemy, Alembic, Pydantic, Spring Data JPA, Hibernate
Databases & Infrastructure : PostgreSQL, Docker, GitHub Actions, Flyway, Maven, Railway, Vercel
Testing & Frontend : pytest, JUnit 5, Testcontainers, React, Next.js, Tailwind CSS, Git

EXPERIENCE

Microsoft Intune (Sponsored MQP) Aug. 2025 – Mar. 2026
Capstone Software Engineer Remote

- Built a Python event-stream pipeline that captures real-time Win32 UI events and converts them into semantic workflow graphs for detecting user friction and UI latency.
- Combined heuristic detectors with a Hidden Markov Model sequence detector (Baum-Welch training, Viterbi decoding) into a single workflow-level detection system, reaching **90.1% F1** (92.8% precision, 87.6% recall) on a human-labeled dataset.
- Cut false positives by **65%** through threshold tuning and failure analysis of dead-click inflation and application-error blind spots.

SerenDPT (Venice Project Center IQP) Mar. 2025 – Jun. 2025
Software Developer Venice, Italy

- Wrote Python and Node.js scripts to collect, clean, and validate infrastructure and rental data across **126 historic islands** for a Venice Project Center study.

PROJECTS

Herd | `herd-wpi.vercel.app` Oct. 2026 – Present

- Built and launched a campus comment site for WPI students (FastAPI, PostgreSQL, Next.js, TypeScript) on Railway and Vercel; **30+ approved student reviews** in the first three days, 24 of them on launch day.
- Built the abuse controls: comments held until approved, pattern checks that sort the review queue, Cloudflare Turnstile on submit, Postgres-backed rate limits, and auto-hide after two reports, with a row lock so simultaneous reports can't miss the threshold.
- Built a **395-test** pytest suite against a real PostgreSQL database and broke guards on purpose to confirm the right tests failed.

Vigil | `getvigilapp.vercel.app` May 2026 – Jul. 2026

- Built a telemetry ingestion API (FastAPI, async SQLAlchemy, PostgreSQL) with z-score, error-rate, and repeated-error detectors that open incidents automatically and stream events to a live dashboard over WebSockets.
- Used a partial unique index to stop concurrent requests from opening duplicate incidents, and consolidated the dashboard's service-health queries into one round-trip with a `ROW_NUMBER` window function.

DropForge | `dropforge-roan.vercel.app` Jun. 2026 – Aug. 2026

- Built a limited-drop reservation and checkout backend (Spring Boot, PostgreSQL, Flyway) with idempotent reservation creation and an atomic conditional inventory update that prevents overselling under concurrent requests; pessimistic row locks serialize checkout and cancellation.
- Wrote **81 automated tests** (24 unit, 57 integration) with JUnit 5 and Testcontainers against real PostgreSQL, run in GitHub Actions CI, including failure-injection tests that confirm a mid-transaction error rolls back the inventory change.